

The Relationships between Student Engagement and the Academic Achievement of First-Year University Students in Cambodia

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Abstract This study examined the relationships between student engagement in academically relevant activities and their academic achievement in the first year of university in Cambodia, and tested the presence of conditional effects of student engagement on achievement by students' gender, geographical origin, and pre-university academic experience. The participants were 919 first-year students at nine universities in Phnom Penh City. Student engagement was measured using a revised version of the National Survey of Student Engagement and the College Student Experiences Questionnaire. The regression analyses revealed that the student engagement in time spent on out-of-class course-related tasks, homework/tasks, and active participation in classroom settings added significant values to Cambodian student achievement. Contrary to the extensive student engagement literature in developed countries, student engagement in out-of-class peer learning and extensive reading did not make any meaningful impacts on student achievement in the present study. Certain effects of student engagement on achievement tended to differ in magnitude by students' pre-university academic experience and geographical origin. Implications for policy are discussed with a call for solid programs to promote and enhance the quality of student engagement activities on-and off-campus, especially among the academically disadvantaged students.

Keywords Academic achievement · Cambodia · Conditional effects · Student engagement · University students

Introduction

For the last few decades, an extensive body of research has provided empirical evidence of the relationships between student engagement in educationally purposeful activities and desired learning outcomes. With few exceptions, the bulk of this study has suggested a positive impact of student engagement on various measures of students' learning outcomes (e.g., Bai and Pan 2009; Carini et al. 2006; Davis and Murrell 1993; Greene et al. 2008; Kuh et al. 2008; Kuh et al. 1997; Laird and Cruce 2009). However, most of this evidence largely pertains to the context of countries with a far more developed higher education systems. Relatively little work has extensively explored the impact of student engagement on achievement in the countries with evolving and troubled higher education systems, especially within the Asian contexts. Thus, as the nature of student learning is not context-free (Kahu 2011), it is of utmost importance that more research be undertaken in order to understand differential influences of student engagement on achievement outside of the context of the existing literature.

Cambodia is a case in point for this present study. In recent years, while a lot of progress has been made toward the understanding of possible predictors of student achievement at the basic education level (e.g., Chhinh 2003; Marshall et al. 2009), the country's struggle to increase the student achievement remains critically underrepresented in higher education (Kwok et al. 2010). Particularly, even with the recent efforts by the Accreditation Committee of Cambodia (ACC) to warrant institutional evaluations so as to assure the quality of higher education (Chet 2006), research on the relationships between student engagement and achievement remains scarce. The current study, thus, sought to address the gaps in the university impact literature and the scholarship of university student learning in Cambodia by providing a new

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empirical outlook on the notion of student engagement and its link with student achievement from a Cambodian cultural perspective. To add insightful evidence to the literature, it also extended the understanding of the complex influences of student engagement on achievement by examining whether the relationships differed in magnitude within specific student populations.

Theoretical Backgrounds

Student engagement is a widely studied and theorized concept in university impact literature. Major advocates of the concept of student engagement are Astin (1984), Pace (1990) and Kuh (2003). Although, these scholars used different terms to describe the student engagement processes, their emphases did not deviate substantially from one another. Astin, for example, broadly characterized student engagement in terms of students' mental and physical involvement in the academic experience, such as involvement in class activities, relations/contact with peers and teachers, and extra-curricular activities. According to Astin, students' physical and psychological involvement in academic activities represents a valuable asset for successful integration and desired learning outcomes. Pace defined student engagement as the quality of efforts, students devote to learning in three core activities (student-faculty contact, cooperation among students, and active learning) that research has shown to be linked with the desired learning outcomes. Following the behavioral perspective, Kuh emphasized five areas of student engagement in academically driven activities during university (level of academic challenge, active and collaborative learning, student-faculty interaction, enriching educational experiences, and supportive campus environment) that are thought to embrace good educational practices. Within this perspective, student engagement is defined not solely as a measure of student effort, but also as the efforts institutions devote to effective educational practices (Kuh et al. 2008). In sum, although, Kuh and his colleagues appeared to offer a variant concept of student engagement by delineating the dual nature of student engagement, the concepts of student engagement guided by Astin and Pace are, by nature, not necessarily limited to students' efforts/involvement. In practical terms, these engagement behaviors are either tacitly or explicitly driven by the institutions' efforts in providing the learning environment conducive for student learning. In that sense, it is justified that the look into student engagement from the students' behavioral view can provide useful information about both students' learning and institutional practices, especially teaching practices that may directly promote student learning and development. As a preliminary study, this present study, therefore, draws

predominantly on a student engagement model with a particular focus on the amount of time and energy students invest in educationally purposeful activities.

Student Engagement and Learning Outcomes

Previous studies have shown that various forms of student engagement (e.g., time on tasks, active and collaborative learning, and teacher–student interaction) are related to desired learning outcomes (e.g., Carini et al. 2006; Davis and Murrell 1993; Greene et al. 2008; Kuh and Hu 2001; Pike and Kuh 2005). A review of literature has, however, indicated that these studies have been limited in two critical respects. First, the literature has shown that student engagement measures were operationally defined differently across studies. While Pike and Kuh (2005), for instance, deconstructed student engagement broadly into social and academic domains and explored their relationships with desired learning outcomes, Davis and Murrell (1993) broke down academic and social efforts into five subscales, such as clubs and organizations, conversation topics, student acquaintances, writing, and course learning. In one instance, student engagement may embrace only active cooperation among students and student-faculty contact (Kuh et al. 1997). In another instance, it incorporates a wide array of activities, including even more comprehensive global measures of student engagement from the National Survey of Student Engagement (NSSE) and the College Student Experiences Questionnaire (CSEQ) (Kuh 2003; Pace 1990). Unlike other researchers, Kuh et al. (2008) characterized student engagement as only a single construct combined from 19 items to represent educationally purposeful activities. While such a combining method is suggestive of dodging concerns about inconsistent reporting of subscales within large constructs (Pike et al. 2011), activities students are thought to be engaged in are not necessarily equally important across settings. Further, even within the same constructs, it is far from clear whether previous researchers used the same number of variables or measurement items and interpreted them in the same fashion from one context to another. Given these measurement discrepancies, it is questionable as to whether previous results can be compared with one another.

Second, recent literature has suggested that the relationships between student engagement and achievement can be somewhat conditional. Many measures of student engagement such as active learning and cooperation among students were, for instance, found to have more substantial effects on the achievement of low-ability students (Carini et al. 2006), students of color (Kuh et al. 2008), and first-generation students (Pascarella et al. 2004). The effect of student engagement was also embedded within students' gender (Bai and Pan 2009). The presence of these

interaction effects in recent literature seems to question the previous results in that the influences of some insignificant variables on achievement could have been masked within certain subsamples when the interaction effects were not considered. Such a methodological gap in literature is likely to pose a serious question when results are to be compared across studies.

Drawing on the presence of these gaps in the literature, it is imperative that the general and conditional effects of locally relevant student engagement constructs be taken into account to reflect more specific features of student learning within a specific social and cultural context. This study, thus, sought to address two questions:

- (1) What types of student engagement are related to the academic achievement of university students in Cambodia after controlling for their background characteristics (gender, delayed enrollment status, multiple university attendance, geographical origin, parents' university experience, weekly expense, academic resources at home, and pre-university academic experience)?
- (2) Do the effects of student engagement on achievement vary in magnitude by students' gender, geographical origin and pre-university academic experience?

Research Methods

Data Source

The present research focused on first-year undergraduate students as it is a crucial transition period when substantial gains in learning occur (Pascarella and Terenzini 2005; Tinto 2007). The data were a multi-institution sample drawn from both public and private universities in Phnom Penh City. The sample included 919 first-year university students majoring in English at nine universities who completed a questionnaire survey in 2012. This sample represented approximately 30 % of the population of first-year undergraduate students at each university. Among the participating universities, private universities were over-represented because very few public higher education institutions offer four-year university-level English programs. This university profile largely reflects the current situation of higher education in Cambodia, where the private sector has played a major role in widening the gateway to university education (Chet 2006).

Measures

Academic Achievement

First-year English test scores served as the indicator of student achievement in this study. The test assessed

learning with reference to the course book contents in three conventional areas (vocabulary, grammar, and reading skills) largely using a multiple-choice testing format in order to ensure the students' familiarity with the test. The test was validated by homeroom and senior teachers and piloted with 80 students at three different universities prior to the actual administration. The overall internal consistency (α) of the actual test was .805.

Student Engagement

Measures of student engagement were based on a self-report questionnaire revised from the NSSE and CSEQ (see Appendix A). The self-report questionnaire is a widely used tool in university impact research and is considered valid and reliable provided that its contents are appropriately designed, clearly worded, or phrased, referred to respondents' recent activities, perceived to be important and familiar to the respondents (Pike et al. 2006; Pascarella and Terenzini 2005). To meet these conditions, the survey items used for this research were modified in a few respects. For instance, only the questions from the NSSE and CSEQ that could be applied to the context of Cambodian higher education learning were incorporated into the survey. These questions were restructured, deconstructed, piloted, and contextualized accordingly to make them fit into current student learning realities at Cambodian universities. The original four-point Likert scale (in the NSSE and CSEQ) was also changed to a six-point Likert scale. This modification was to ensure that broader response options were given to students so that more objective ratings could be obtained.

The questionnaire consisted of 26 items that tapped student engagement in educationally related activities. These items measured students' study time outside class with a particular focus on time spent on course-related reading (two items), homework (two items), and extensive reading (three items) and their engagement level in active and collaborative learning activities (19 items) using a six-point Likert scale ($6 = \text{always}$; $5 = \text{usually}$; $4 = \text{often}$; $3 = \text{sometimes}$; $2 = \text{rarely}$; $1 = \text{never}$). Measures of time spent on course-related reading and homework were combined into a single variable "*time on course-related tasks*" due to a relatively high collinearity between the two variables ($r = 0.64$, $p < .001$). Factor analysis was conducted on the measures of active and collaborative learning. The analysis produced four distinct factors with two items removed to ensure internal reliability of the scales: peer learning (four items), whole class active participation (five items), reading and class preparation (four items), and frequency of doing homework/tasks (four items). Since these four factors are latent variables loaded from the factor analysis, reliability testing using a Cronbach's alpha (α) as

a criterion was necessary to determine their internal consistency (see [Appendix B](#)). Measures of extra-curricular activities were not included in this study because most of the prescribed items encompass excessive out-of-class encounters between and among students, the practices apparently being at the infancy stage in the Cambodian higher education context. In total, six student engagement variables were obtained and used for the analysis.

Student Characteristics

Student characteristics included gender, delayed enrollment status, multiple university attendance, geographical origin, parents' university experience, weekly expense, academic resources at home, and pre-university academic experience. Delayed enrollment status was obtained by subtracting the high-school graduation year from the enrollment year to university, with a result of "0" denoting "no delay" to the university. Multiple university attendance and geographical origins were dichotomous variables respectively representing whether students are doing two degrees at different universities at the same time and are originally from the city or the provinces. Although, parental education, occupation, and income have been widely used as a proxy for students' family socio-economic status, only parents' university experience, academic resources at home (desk, personal room, study place, computer, the internet, and dictionary) and students' weekly expense were measured since they were considered a more direct proxy for students' social and financial capital that could be obtained with the use of a self-report questionnaire. Pre-university academic experience was represented by standardizing and summing three academic-related measures representing students' academic readiness: the number of years of learning English full-time, the number of years of learning English part-time, and their achieved level of English prior to entering university ranging from the pre-intermediate to the advanced level. The use of Z-scores is more robust in which each item is measured on the same scale, thereby, cancelling out any differences in the measurement unit of each original item when all the items are combined (Kuh et al. 1997). A summary of variables is provided in Table 1.

Data Analysis

Data were analyzed in two phases utilizing blocked, stepwise regression methods. A Predictive Analytics Software (The PASW Statistics 18 for Windows, formerly SPSS Statistics) was used for the analysis. While the use of the stepwise method in regression analysis is limited in that only statistically significant results that meet the established mathematical criteria are reported, and there is also a

possibility of leaving out important predictors, Field (2009) did not rule out its robustness in the exploratory model building. This method was, thus, appropriate for this exploratory research. In the data analysis process, student background characteristics, student engagement, and cross-product variables were entered into three different blocks, respectively, (see Table 2). In the first phase, two general effects regression models were estimated to answer the first research question. Model 1 estimated the effects of student background characteristics (block 1) which included gender, delayed enrollment status, multiple university attendance, geographical origin, parents' university experience, weekly expense, academic resources at home, and pre-university academic experience on students' academic achievement. This model served as a statistical control method to more accurately isolate its differential effects from the subsequent models. Model 2 tested the general effects of student engagement (time on extensive reading, time on course-related tasks, peer learning, whole class active participation, reading and class preparation, and homework/tasks) on achievement by controlling for the confounding effects of student background characteristics. In this model, variables in blocks 1 and 2 were simultaneously evaluated in relation to the achievement.

In the second phase of the analysis, 18 cross-product terms among all the measures of student engagement, students' gender, geographical origin, and pre-university academic experience were created and entered into block 3. These variables were then added to the general effects equation in the second model to test for the presence of conditional or interaction effects (Model 3) and to provide an answer to the second research question. The presence of conditional effects in the model implies that the influence of student engagement on academic achievement varied in magnitude by particular student background characteristics (Kuh et al. 2008). To ease the interpretations of interaction effects, all the scale variables were recoded into dichotomous variables in this phase (0 = below mean; 1 = above mean). Taking into account, the interaction effects could reduce the risk of accepting the main effects of student engagement only on the surface while underestimating its overall effects on student achievement (Pascarella and Terenzini 2005).

In the model development, a few assumptions of the regression model were examined. First, as a rule, when a large number of predictors are examined, there is a possibility that multicollinearity can distort the accuracy of the model estimation. To address this issue, two important statistics (the tolerance statistics and variance inflation factor) were tested. The data indicated that the tolerance statistics and variance inflation factor were far above 0.2 and <10, respectively. Thus, as Menard (1995) and Myers (1990) suggested, multicollinearity within the data was not

Table 1 Description of variables

Measure	Description/Metric	N	Mean	SD
Academic achievement	English test score (total score = 76)		39.39	9.51
<i>Student characteristics</i>				
Gender	0 = male	529 (57.6 %)		
	1 = female	390 (42.4 %)		
Delayed enrollment to university	0 = not delayed	477 (51.9 %)		
	1 = delayed	442 (48.1 %)		
Multiple university attendance	0 = no	516 (56.2 %)		
	1 = yes	403 (43.8 %)		
Geographical origin	0 = province	458 (49.8 %)		
	1 = Phnom Penh	461 (50.2 %)		
Mother's university experience	0 = no	803 (87.4 %)		
	1 = yes	116 (12.6 %)		
Father's university experience	0 = no	693 (75.4 %)		
	1 = yes	226 (24.6 %)		
Student's weekly expense	Average weekly expense (in Riels ^a)		79,893	67,743
Academic resources at home	Academic facilities at home ^b (0–6)		3.91	1.53
Pre-university academic experience	Mean of three z-scored items		−0.00	0.67
<i>Student engagement</i>				
Time on extensive reading	Hours a week on extensive reading		1.98	2.25
Time on course-related tasks	Hours a week on readings and homework outside class		2.69	2.39
Peer learning	Frequency of discussion with peers		3.85	1.00
Whole class active participation	Frequency of whole class participation		4.15	0.92
Reading and class preparation	Frequency of reading and class preparation		3.77	1.01
Homework/tasks	Frequency of doing homework/tasks		4.78	0.98

^a 'Riel' is the currency used in Cambodia

^b Availability of a desk, personal room, study place, a computer, the internet, and a dictionary at home

a major concern (as cited in Field 2009). Second, the fact that student's weekly expense ($M = 79.893$; $SD = 67.743$) was positively skewed, as evidenced in its extremely large standard deviation, might in principle have biased the regression models significantly (see Table 1). The influence of some extreme cases on the whole model is acknowledged in Field (2009). One solution is to remove those cases to improve the goodness-of-fit of the model. But, the fact that Cook's distance was reported <1 in this study suggests that the outliers did not make any substantial changes in the regression models even when the models were regressed in a reduced form (Cook and Weisberg 1982, as cited in Field, 2009). Thus, no removal of extreme cases by student's weekly expense was made in the analysis. Finally, assumptions of linearity, homogeneity of variance, and normality of residuals were evaluated to determine whether the findings of this study could be generalized to a wider population using a plotted graph

between regression standardized predicted values (ZPRED) and regression standardized residuals (ZRESID), a histogram and normal probability plot of the residuals. Field (2009) suggested that these assumptions are met when the graph of ZPRED and ZRESID does not funnel out and consist of any sort of curve, the residuals are normally distributed having a bell-shaped curve, and the points representing the regression standardized residuals lie on a straight line. The analysis of residuals indicated that this study met these conditions, thus, implying that the results from the regression model could be generalized beyond the sample.

Results

The first regression model estimated the effects of student characteristics on achievement. This model accounted for

Table 2 Regression coefficients of the significant predictors of academic achievement

Variable	Model 1	Model 2	Model 3
<i>Block 1^a (Student characteristics)</i>			
Pre-university academic experience	4.002 (.466)***	3.165 (.428)***	1.546 (.507)**
Academic resources at home	1.105 (.202)***	.872 (.186)***	.715 (.184)***
Delayed enrollment	1.770 (.583)**	1.838 (.532)**	1.877 (.520)***
<i>Block 2^b (Student engagement)</i>			
Time on course-related tasks		1.241 (.118)***	.834 (.133)***
Homework/tasks		1.964 (.282)***	1.903 (.276)***
Whole class active participation		1.343 (.277)***	1.448 (.271)***
Time on extensive reading		−.341 (.126)**	−.338 (.123)**
<i>Block 3^c (Interaction terms)</i>			
Time on course-related tasks × geographical origin			.699 (.150)***
Reading and class preparation × pre-university academic experience			1.424 (.317)***

The results were based on blocked, stepwise regression analyses

R^2 for model 1 = .148 (adj. R^2 = .146); R^2 for model 2 = .302 (adj. R^2 = .297); R^2 for model 3 = .334 (adj. R^2 = .327) (** p < .01; *** p < .001); Standard errors are in parentheses

^a Block 1 includes gender, delayed enrollment status, multiple university attendance, geographical origin, parents' university experience, weekly expense, academic resources at home and pre-university academic experience

^b Block 2 includes time spent on extensive reading, time spent on course-related tasks, peer learning, whole class active participation, reading and class preparation and homework/tasks

^c Block 3 includes 18 cross-product variables representing the interaction terms of six student engagement factors in block 2 by gender, geographical origin and pre-university academic experience

14.8 % of the variance in achievement, $F(3,915) = 53.11$, $p < 0.001$. The results showed that students from different backgrounds differed significantly in their achievement. As shown in model 1 (Table 2), pre-university academic experience had the strongest and positive impact on achievement. Students with a higher level of academic experience before entering university tended to perform better than their counterparts. The stepwise regression indicated that student achievement was not significantly related to gender and geographical origin of students. Additionally, this study did not find any potential effects of parents' education and students' weekly expense on achievement. It was the academic resources at home that exerted a positive influence on achievement. The results also revealed that delayed enrollment to university positively affected achievement. Students entering university immediately after high-school graduation appeared to be low-performing compared to those who entered the university later. Interestingly, multiple university attendance was not found to affect achievement in this study.

Model 2 estimated the effects of student engagement on academic achievement. By statistically controlling for student characteristics, student engagement in educationally purposeful activities added significant values to student academic performance, explaining an additional 15.4 % of the variance in achievement, $F(7,911) = 56.41$, $p < 0.001$. That is, although, students' pre-university academic experience, academic resources at home, and

delayed enrollment status continued to exert some preexisting influences on student learning, some forms of student engagement clearly made an additional impact. The results showed that the amount of time spent on course-related tasks, frequency of doing homework/tasks, and whole class active participation positively influenced the student achievement. Surprisingly, peer learning did not make any impact on student achievement. The relationship was even negative despite being insignificant ($B = -0.009$, $p = 0.75$). The regression results further showed that the time spent on extensive reading was negatively related to achievement, suggesting that an increase in time spent on reading English for purposes other than those related to the coursework (extensive reading) did not exert any meaningful influence on student achievement.

Model 3 took into account the interactions between student engagement in academic-related activities and achievement based on students' gender, geographical origin, and pre-university academic experience. By adding these cross-product terms to the full model, variability in student achievement was improved significantly from 30.2 to 33.4 %, $F(9,909) = 53.30$, $p < 0.001$. This implies that the general effects model could be masked given that the interaction terms were not taken in account. As presented in Table 2, certain forms of student engagement varied in the effect on achievement based on students' geographical origin and pre-university academic experience, whereas the effects were not found to be restricted to gender difference.

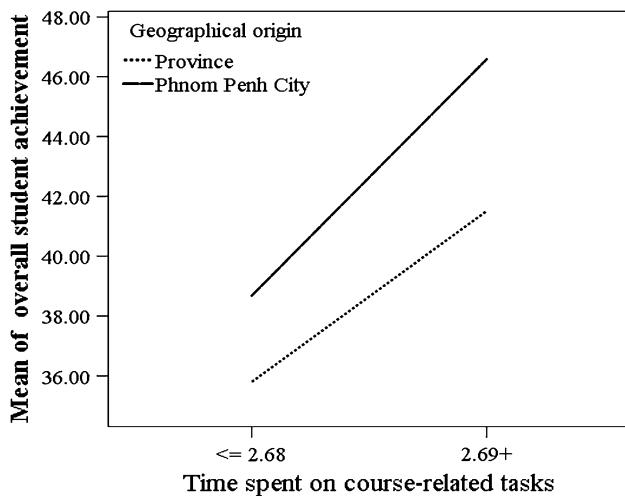


Fig. 1 Effect of time spent on the course-related tasks on achievement by geographical origin

Findings in model 3 showed that the time spent on course-related tasks outside the class had both general and conditional positive effects on achievement, with a greater effect for students in the city. While the gap in academic performance between the two groups was not very pronounced given that both type of students spent below average time on academic tasks outside the class ($M = 2.69$ h/week), students from Phnom Penh City tended to be the far better group when their study time increased, although, engaging in these out-of-class activities also had a moderate compensatory effect on students from the disadvantaged areas, i.e., from the provinces (see Fig. 1).

The results further indicated that reading and class preparation had an interaction effect on the academic achievement of students with different pre-university academic experiences. The presence of the interaction effect of reading and class preparation on student achievement unveiled a blurring effect of this variable when group difference was not considered. In the general model, student achievement was not significantly associated with students' effort in reading and class preparation. Yet, when pre-university academic experience was allowed to interact with the students' effort in class preparation, this cross-product effect began to exert a significant and positive influence on achievement. This result indicated that an increasing amount of reading and class preparation per se did not necessarily translate into productive learning outcomes. In contrast, students' pre-university academic experience concurrently played a mediating role in fostering the desired learning outcomes. As Fig. 2 suggests, the positive and large effect of student engagement in reading and class preparation was more striking for students with a higher pre-university academic profile. This result suggests that the effect of student engagement in

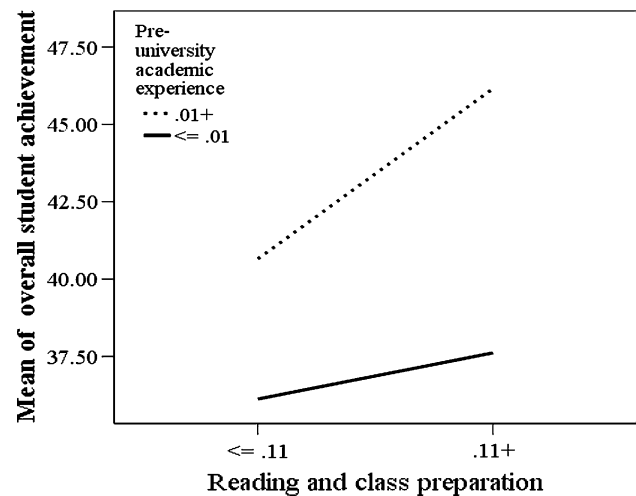


Fig. 2 Effect of reading and class preparation on achievement by pre-university academic experience

reading and class preparation on achievement was more conditional, with greater advantages for students possessing a higher academic experience prior to entering the university.

Limitations

The results of this study are limited in some respects. First, although, the sample was multi-institutional, this study failed to account for data representing students from other popular programs such as business, economics, and information technology and from non-urban universities. Thus, these results may not reflect the Cambodian university students' learning situations in its entirety. Second, given that student engagement was measured at a single moment time on first-year students, changes in how students exhibit their learning behaviors over time cannot be detected. The effects of student engagement found in this study may have differed to some extent, had this study accounted for longitudinal data that track student engagement over time. It is advisable that a longitudinal study in a similar setting be conducted. Third, this study reported student engagement purely using a self-report questionnaire. Although, the self-report questionnaire, when systematically designed, is robust to reflect on student learning from the students' perspective, students' attitude toward the rating of lengthy question items may change at times. Further research using qualitative data is needed to gain greater insight into effective learning practices at Cambodian universities. Finally, although, the researcher tried to elaborate on student engagement constructs by breaking them down into more contextually comprehensible subsets, student engagement in this study does not necessarily embrace all the commonly documented process indicators in the

university impact literature. Student engagement was wholly constrained in meaning to the students' learning behaviors.

Discussion

This study employed a student engagement model to examine the relationships between student engagement in educationally purposeful activities and their academic achievement in both general and conditional terms. Based on the regression analyses, the amount of time and energy Cambodian students invested in educationally driven activities explained about 20 % of the variance in their academic achievement. Though small in magnitude, this effect is relatively meaningful when compared with that of the existing student engagement literature (e.g., Carini et al. 2006; Kuh et al. 2008) considering the nature of the present study which did not account for other highly documented indicators of student engagement—student–teacher interaction, extra-curricular activities, and non-academic activities, the dual nature of student engagement which includes other important influences of institutional practices and, more importantly, classroom effects. This result seemingly provides a meaningful knowledge base for policy makers, educators, and researchers to improve their understanding of what student engagement factors make a difference in student learning in Cambodia and what do not. Overall, three sets of findings emerged from this study. First, results of this study have provided some important insights into various engagement activities that are meaningful for student learning. This study identified three main dimensions of student engagement that had general and positive payoffs in student achievement: time spent on course-related tasks outside the class, frequency of doing homework/academic tasks, and whole class active participation.

The significant effects of the time students devoted to course-related tasks outside the class and the frequency of doing homework/academic tasks on student achievement highlight the importance of extra-hour study among first-year students in the context of this study in preparation for academic challenges at the university level. One possible reason is that, in a coursework-based learning environment like Cambodia's, course-related tasks are considered a widely employed tool by teachers to follow-up the extent to which the students are supposed to have achieved after a certain period of time. Thus, commitment to these activities put students at an advantage. These results seem to be consistent with that of recent student engagement literature (e.g., Carini et al. 2006; Kuh et al. 1997; Kuh et al. 2008). Meanwhile, whole class active participation was positively associated with student achievement. From a theoretical standpoint, this result delineates the importance of

classroom involvement on the part of students, the attribute largely supporting Astin's (1984) involvement paradigm. From a practical standpoint, class participation represents great willingness and motivation that students put forth to capitalize on the restricted availability of a productive time with peers and particularly with teachers via in-class encounters as even the unofficial out-of-class encounter time with teachers remains limited in the Cambodian higher education context (Chen et al. 2007). Thus, the utilization of classroom settings in the Cambodian context can be one of the most fruitful options for increasing students' learning opportunities.

Second, the effects of certain engagement activities were not as meaningful as they should have been. The present study found that time on extensive reading was negatively related to achievement. This is surprising as extensive reading would prove useful for students' learning in many respects such as reading skills and language development. However, the lack of reading skills among Cambodian students may have prevented them from fully capitalizing on such additional reading activities because Cambodian students tend to lack reading habits even in the Cambodian language (Pit and Ford 2004). Similarly, the effect of peer learning was also found in a reverse direction, though statistically insignificant. There are two possible explanations for this contradictory result. First, peer learning may be relatively new for first-year students in Cambodia as they may be in a transition period from high school to university. Students may have been in the process of integrating themselves into the university-level learning context and especially reconciling between the traditional learning modality by which they had been influenced at the lower education level (Neau 2003) and a newer approach at the university that tends to promote independent learning. Working together as a group may, thus, be a real challenge for them at this stage. Benefits from peer collaboration may materialize when they become fully accustomed to this group learning modality. Second, engagement in peer learning in the context of this study may instead denote academic difficulty students were already facing, but striving to handle through seeking helps from their peers since two of the four items of peer learning reflect help-seeking behaviors—"asked for help from friends when having (a) learning problem(s)" and "had discussions with other students on learning difficulties". This would potentially result in struggling students scoring higher on this variable and, thus, explain the negative relationship with their achievement.

Finally, the effects of student engagement on achievement are in some ways conditional. Contrary to studies in the US or the Western context where engagement reportedly had compensatory effects on students from disadvantaged backgrounds (e.g., Carini et al. 2006; Kuh et al. 2008; Pascarella et al. 2004), the effect of time spent on course-related tasks outside the class was especially greater

for students residing in Phnom Penh City. These urban students benefited more substantially from this form of academic engagement than their non-urban counterparts. This conditional effect may reflect preexisting advantages, at least in terms of pre-university academic experience, with which the former group were equipped prior to university entry. One possible explanation is that students in the city may enjoy more access to diverse learning environments and higher quality education than those in the provinces and, thus, be more likely to obtain a richer academic experience to help prepare for challenges at the university. This holds true given the fact that the educational development within the country is far more concentrated in the city (UNESCO 2011). As a result, although, students of the two populations spent a similar amount of time on self-learning outside the class, the abilities to effectively translate such endeavors into productive ends may have been different, with non-urban students likely to be an inferior group despite also having seen some progress in achievement.

Meanwhile, the positive effect of reading and class preparation on achievement was more pronounced for students with a higher pre-university academic profile. This finding should come as no surprise since the quality of reading and class preparation may have been mediated by students' prior academic experience. One possible reason is that students with a higher academic profile may possess better academic preparation prior to university in terms of both background knowledge and study skills. Conversely, students who attended the university with lower academic experience may lack prior academic preparation, a factor that is common among students in many Asian countries, including Cambodia (Chapman 2009). These students might be in a trial and error process to search for appropriate learning strategies to compensate for their lack of learning experience. This is quite true in a country like Cambodia where the provision of preparatory courses before and/or after the university entry and/or comprehensive course orientations by the universities is not fully in place. Thus, coupled with academic difficulty at the university, the progress made by these students may not be substantial during the first-year period. The gains may be more robust when students become accustomed to the university-level challenges.

Conclusions and Implications

The findings from this study provide significant evidence to a growing body of research on student engagement at the university and its relation to their academic achievement. Although, recent studies in the US and Western settings have largely supported the collective power of overall student engagement in educationally relevant activities,

this study concluded that student engagement in the form of time spent on course-related tasks outside the classroom, homework/tasks, and whole class active participation were the stronger predictors of student achievement; whereas peer learning did not exert differential influences on first-year university students' academic achievement. Results also indicated that extensive reading activities had a negative relationship with student achievement. Certain features of engagement (i.e., time spent on course-related tasks and reading and class preparation) were found to have greater effects on students in the city and those with a higher level of pre-university academic experience, respectively. Although, these interaction effects seemed to contrast with the previous research in that engagement efforts were quite fruitful for students from disadvantaged backgrounds, the examination of conditional effects of student engagement on achievement is vitally important so as to reveal the blurring effect of some important variables (e.g., reading and class preparation) that would have been masked within specific student populations had they been tested only in the general effects model.

Two major implications can be drawn from this study. First, this study offers support for the idea that success in learning, at least in the first year of university, depends on student engagement in academic activities both outside and inside class. One clear implication of this evidence is the need to extend these forms of practice in a more frequent and systematic manner. Schools should rigidly make these activities one of the main parts of their assessment protocols to promote student engagement both on- and off-campus. Specifically, since mostly educationally disadvantaged students made less gain in achievement from their engagement, it is recommended that a tracking system be put in place to provide academic support for this group. More importantly, given that the effect of engagement in literature is usually larger for students with a low-academic profile, bridging programs to enhance their study and reading skills should be made available at this early stage of university to allow this group to catch up with their counterparts. Second, while cooperation among peers and extensive reading are theoretically important for university students majoring in language, the negative effects of these two engagement activities, in turn, highlight the importance of the quality, not necessarily the quantity, of practice among students. Because peer learning and extensive reading culture are new and perhaps difficult for low-ability students, the approaches that schools or teachers have long been introducing to students should perhaps be reconsidered. Extra-curricular programs that promote reading culture and study skills should be emphasized so as to enable the students to fully capitalize on this form of student-centered approach at Cambodian universities.

Appendix A

The table below entails all student engagement indicators used in the survey. Measures of student engagement in educationally purposeful activities were revised from the NSSE and CSEQ.

Original NSSE and CSEQ items (<i>Very often</i> = 4; <i>Often</i> = 3; <i>Occasionally/Sometimes</i> = 2; <i>Never</i> = 1)	Modified questionnaire items for survey (<i>6 = always</i> ; <i>5 = usually</i> ; <i>4 = often</i> ; <i>3 = sometimes</i> ; <i>2 = rarely</i> ; <i>1 = never</i>).
<i>Active and collaborative learning</i>	
Came to class without completing readings or assignments.	Came to class without completing homework. Came to class without completing the assigned reading tasks.
Revised a paper or composition two or more times before you were satisfied with it.	Turned in homework late. Turned in homework with poor quality.
Did additional readings on topics introduced and discussed in class.	Did additional readings on topics introduced in class. Read new materials as a preparation for the next class.
Summarized major points and information in your readings or notes.	Summarized information from your readings.
Made outlines from class notes or readings.	Took detailed notes from class.
Asked questions in class or contributed to class discussions.	Contributed ideas to whole class discussion. Asked questions in class when you don't understand.
Made a class presentation.	Made a class presentation from your group work.
Used a dictionary or thesaurus to look up the proper meaning of words.	Used a dictionary to search for the meaning of new words during class. Used a dictionary to search for the meaning of new words before class.
Worked with other students on projects during class.	Worked actively with other students on the assigned task(s) in small group activities in class.
Discussed ideas from your readings or classes with others outside of class (students, family members, co-workers etc.).	Discussed ideas from your readings or classes with other students outside class.
Sought out a friend to help you with a personal problem.	Asked for help from friends when having (a) learning problem(s).
Tutored or taught other students.	Taught other students.

Appendix continued

Original NSSE and CSEQ items (<i>Very often</i> = 4; <i>Often</i> = 3; <i>Occasionally/Sometimes</i> = 2; <i>Never</i> = 1)	Modified questionnaire items for survey (<i>6 = always</i> ; <i>5 = usually</i> ; <i>4 = often</i> ; <i>3 = sometimes</i> ; <i>2 = rarely</i> ; <i>1 = never</i>).
Been in a group where each person, including yourself, talked about his or her personal problems.	Had reviews of your performance on homework or quizzes with other students. Had discussions with other students on learning difficulties.
<i>Time on course-related reading</i>	
Number of hours per week spent on preparing for class	Hours per week spent reading course-related materials at home Hours per week spent reading course-related materials at school/library
<i>Time on homework</i>	
Number of hours per week spent on writing papers/reports	Hours per week spent doing assigned homework at school/library Hours per week spent doing assigned homework at home
<i>Time on extensive reading</i>	
Number of hours per week spent on other activities related to academic programs	Hours per week spent reading books in English Hours per week spent reading articles or news in English on the internet Hours per week spent reading magazines or newspaper in English

Appendix B

A list below describes student engagement indicators used for the data analysis.

Peer learning ($\alpha = .75$)

- Discussed ideas from your readings or classes with other students outside class.
- Asked for help from friends when having (a) learning problem(s).
- Had reviews of your performance on homework or quizzes with other students.
- Had discussions with other students on learning difficulties.

Whole class active participation ($\alpha = .70$)

- Made a class presentation from your group work.
- Contributed ideas to whole class discussion.
- Worked actively with other students on the assigned task(s) in small group activities in class.
- Asked questions in class when you don't understand.

Appendix continued

Used a dictionary to search for the meaning of new words during class.

Reading and class preparation ($\alpha = .73$)

Read new materials as a preparation for the next class.

Did additional readings on topics introduced in class.

Summarized information from your reading.

Took detailed notes from class.

Homework/tasks ($\alpha = .68$)

Turned in homework with poor quality.

Turned in homework late.

Came to class without completing homework.

Came to class without completing the assigned reading tasks.

(These negative items were reverse-coded.)

Time on course-related tasks

Hours per week spent reading course-related materials at home

Hours per week spent reading course-related materials at school/library

Hours per week spent doing assigned homework at home

Hours per week spent doing assigned homework at school/library

Time on extensive reading

Hours per week spent reading books in English

Hours per week spent reading articles or news in English on the internet

Hours per week spent reading magazines or newspaper in English

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